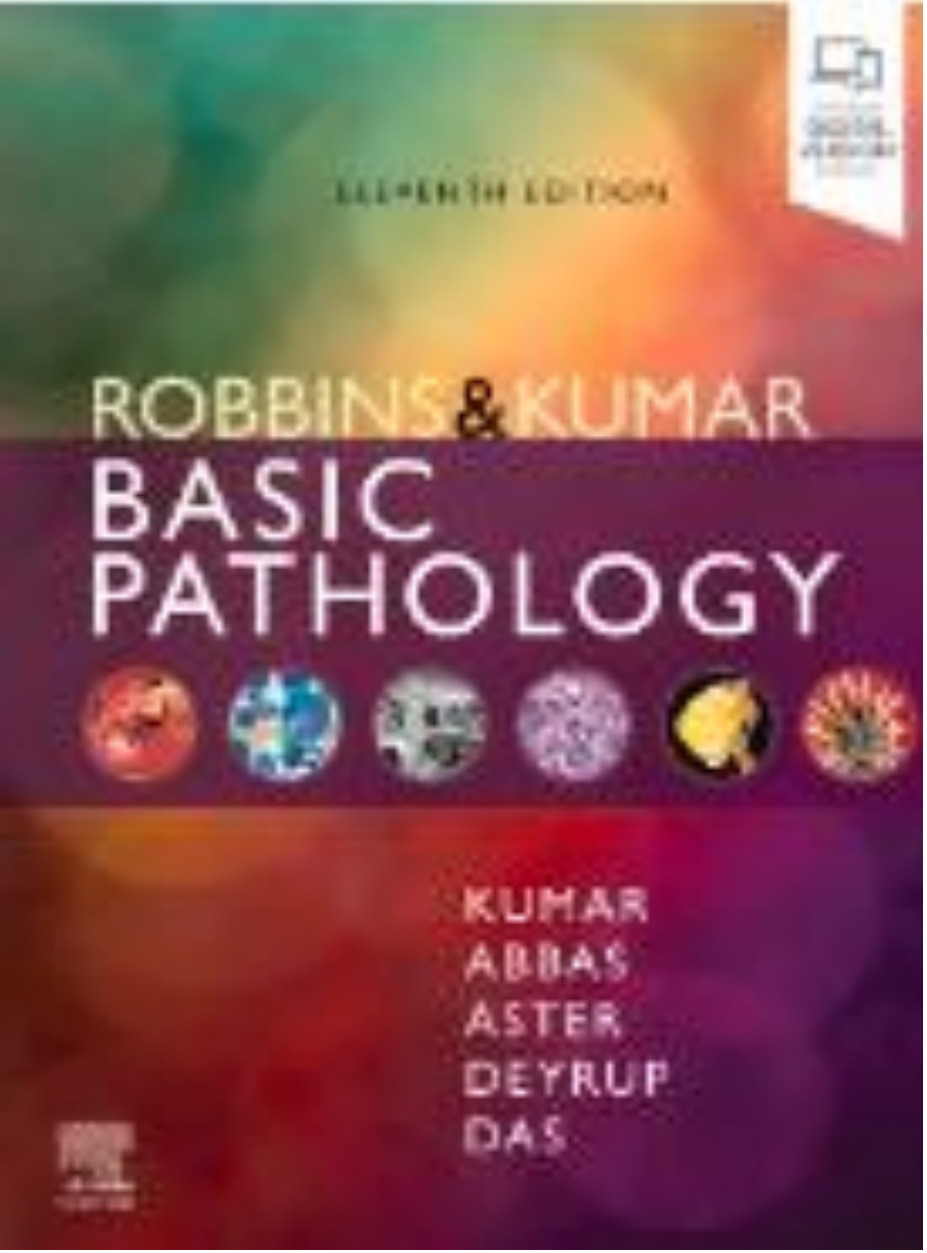


Epilepsy

الاء محمد الخوتاني
M.B.B.S, FRCPC
Neuropathologist
Associate Professor
Pathology Department


ROBBINS & KUMAR
BASIC PATHOLOGY
KUMAR
ABBAS
ASTER
DEYRUP
DAS


NO PHOTOGRAPHY
NO VIDEO RECORDING

Learning Objectives

- Define epilepsy
- Discuss the most common epilepsy-associated diseases and their pathological changes
- Describe the pathological brain changes associated with long-term epilepsy

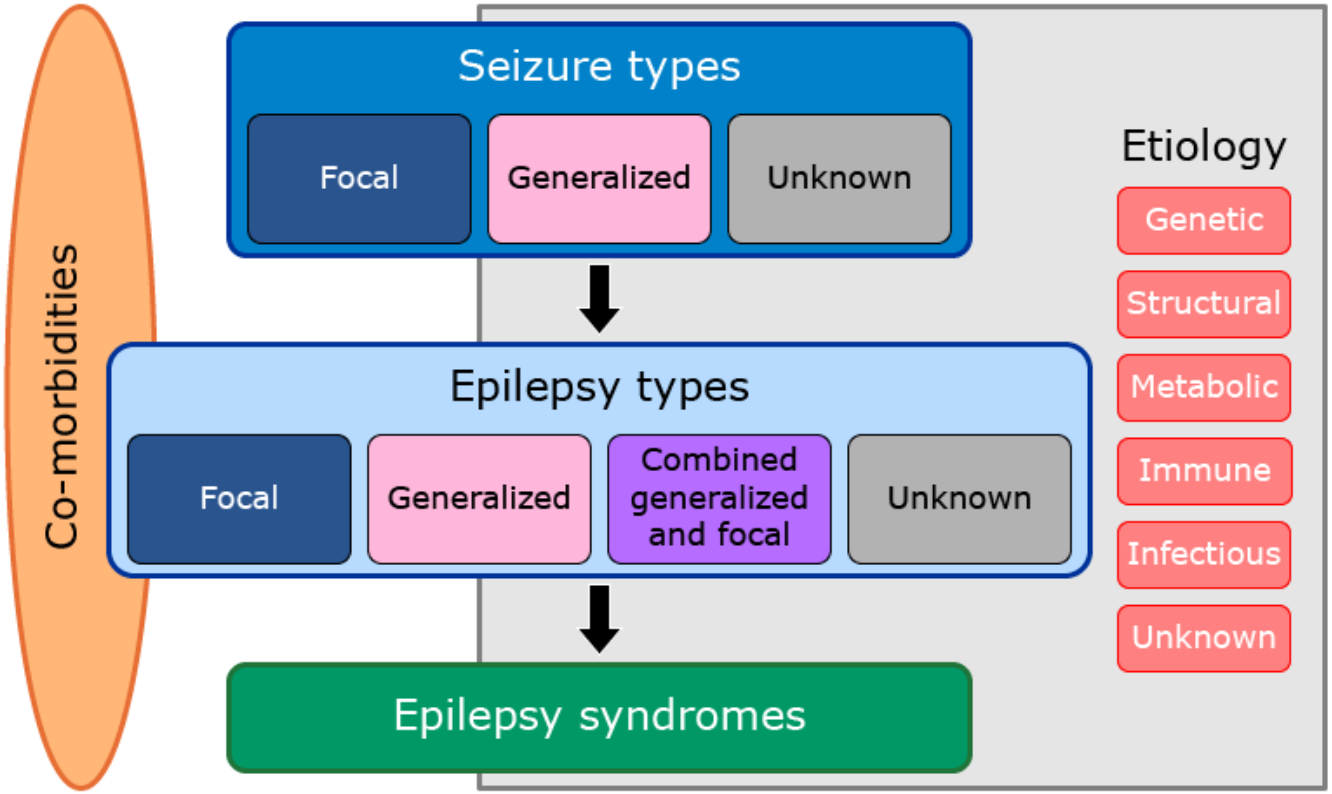
Epilepsy

- A disorder of the brain characterized by an enduring predisposition to epileptic seizures
- A heterogeneous condition characterized by multiple possible seizure types and syndromes, diverse etiologies, and variable prognoses

Updated definition - official revision

A disorder of the brain characterized by a predisposition to recurrent, unprovoked seizures due to abnormal electrical activity in the brain

International League Against Epilepsy (ILAE) framework for seizure and epilepsy classification



From: Scheffer IE, Berkovic S, Capovila G, et al. ILAE classification of the epilepsies: Position paper of the ILAE Commission for Classification and Terminology. Epilepsia 2017. DOI: 10.1111/epi.13709. http://onlinelibrary.wiley.com/wol1/doi/10.1111/epi.13709/abstract. Copyright © 2017 The International League Against Epilepsy. Reproduced with permission of John Wiley & Sons Inc. This image has been provided by or is owned by Wiley. Further permission is needed before it can be downloaded to PowerPoint, printed, shared or emailed. Please contact Wiley's permissions department either via email: permissions@wiley.com or use the RightsLink service by clicking on the 'Request Permission' link accompanying this article on Wiley Online Library (http://onlinelibrary.wiley.com).

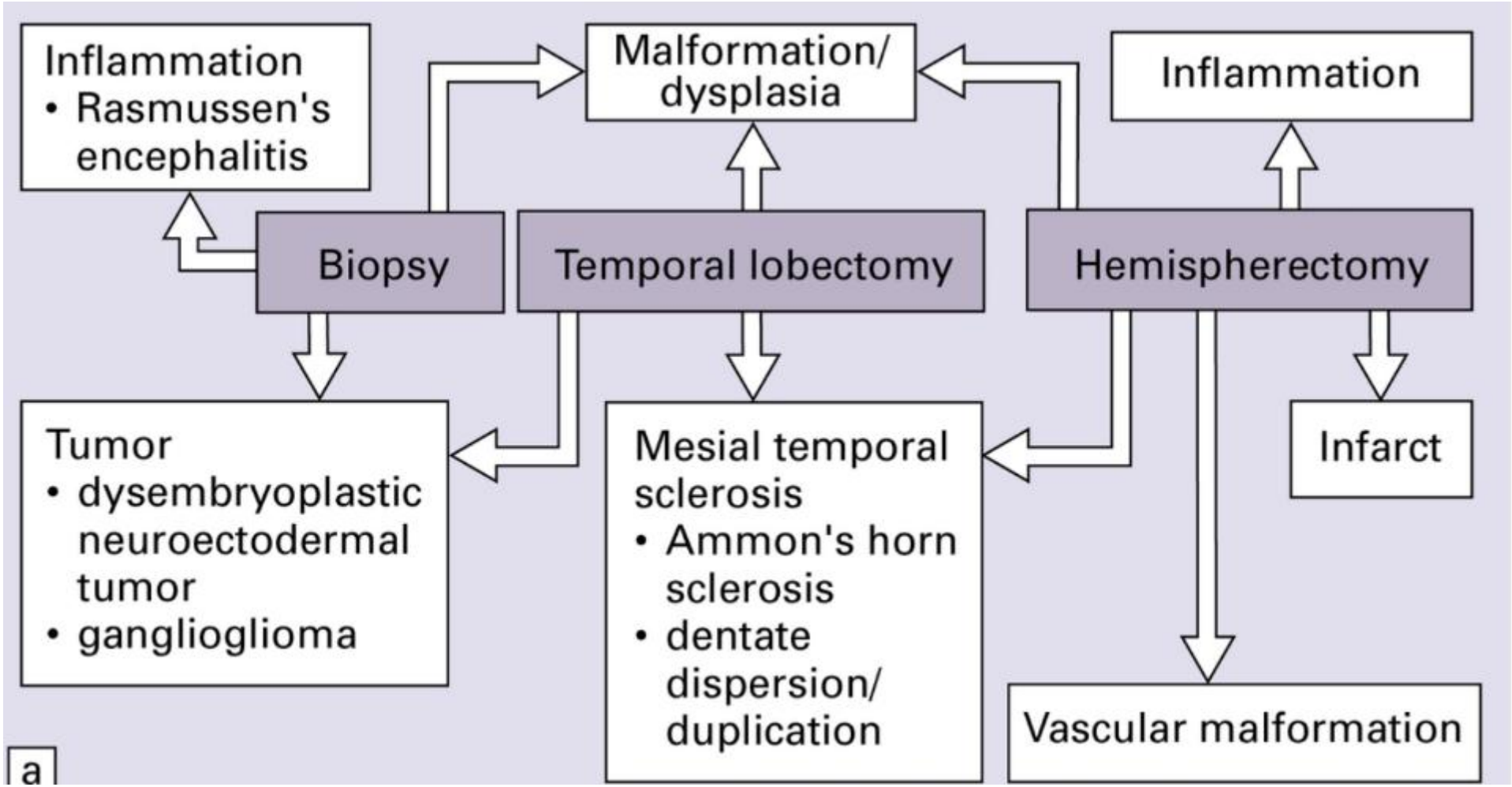
UpToDate®

Etiology

1. Genetics:
   - Idiopathic generalized epilepsies,, childhood and juvenile absence epilepsy, juvenile myoclonic epilepsy, and generalized tonic-clonic seizures
2. Structural:
   - Congenital (cortical dysplasia, tuberous sclerosis)
   - Acquired (stroke, trauma, infection, mesial temporal sclerosis, and tumours)
3. Metabolic: creatine deficiency syndromes, mitochondrial cytopathies and others

Etiology

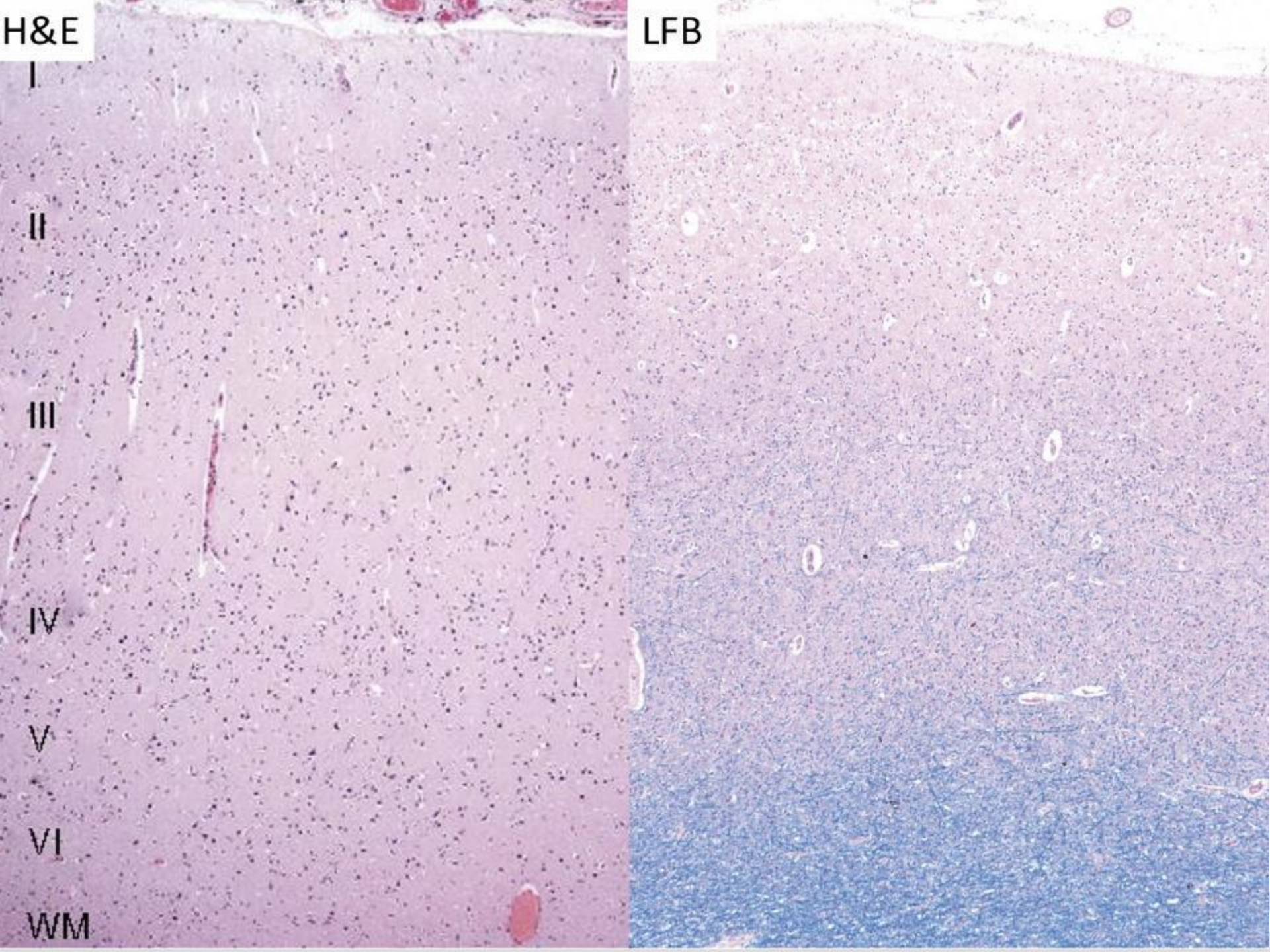
4. Immune:
   - Rasmussen encephalitis, anti-N-methyl-D-aspartate (NMDA) receptor encephalitis paraneoplastic encephalitis
5. Infectious: both acute symptomatic seizures as well as epilepsy
   - HIV, neurocysticercosis, onchocerciasis, malaria, and tuberculosis.
6. Unknown



https://bookshelf.health.elsevier.com/reader/books/9780702057687/epubcfi/6/40[%3Bvnd.vst.idref%3DaB9780723435150000076]!/4/2/4/8/2/6/5:327[ter%2Cs.]

Normal Cortex

- Isocortex (neocortex)
- Allocortex:
  - Archicortex
  - Paleocortex


H&E
LFB
I
II
III
IV
V
VI
WM



Focal Cortical Dysplasia

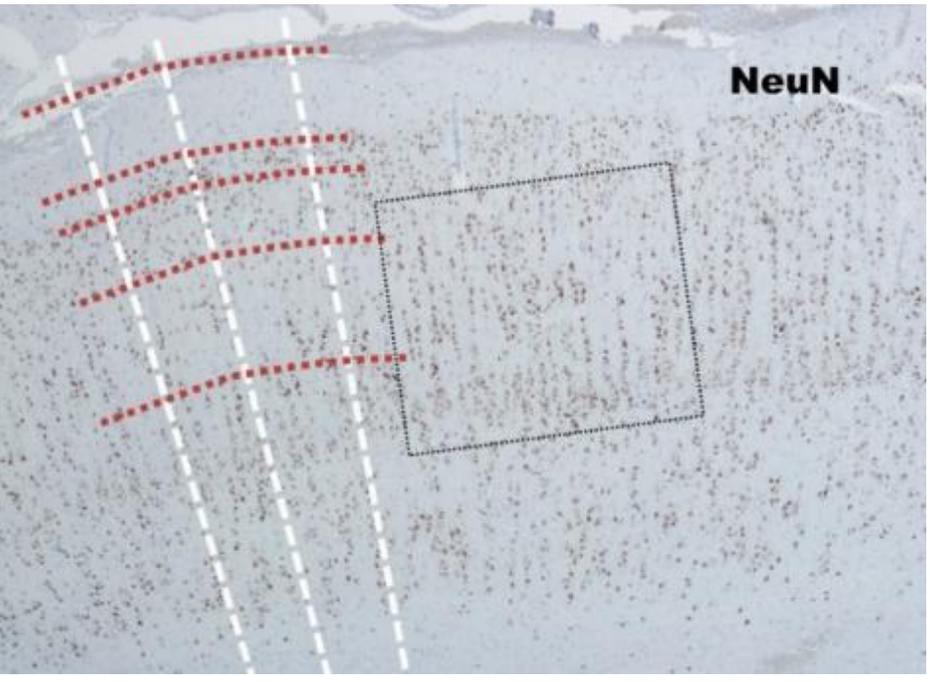
- Heterogeneous group of disorders of cortical formation, which may demonstrate both architectural and proliferative features.
- One of the most common causes of drug resistant epilepsy

Focal Cortical Dysplasia

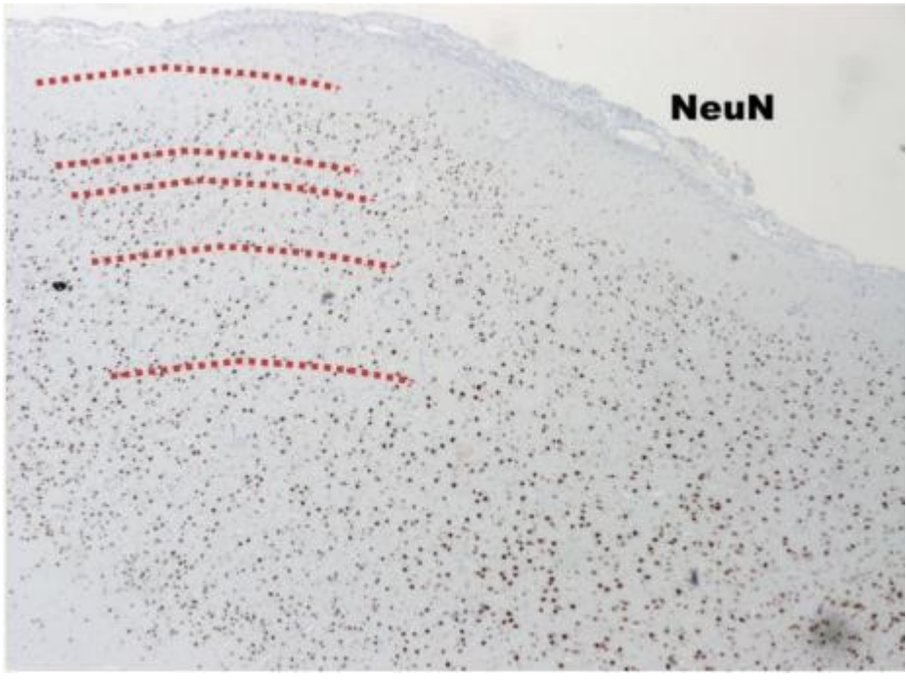
FCDI [a]	FCDIa abundant microcolumns	FCDIb abnormal layering	FCDIc vertical and horizontal abnormalities	
FCDII [a]	FCDIIa dysmorphic neurons		FCDIIb dysmorphic neurons and balloon cells	
FCDIII [a]	FCDIIIa cortical dyslamination associated with hippocampal sclerosis	FCDIIIb cortical dyslamination adjacent to brain tumor	FCDIIIc cortical dyslamination adjacent to vascular malformation	FCDIIId cortical dyslamination adjacent to lesion acquired during early life, e.g. stroke
White Matter [a]	mMCD [b] with excessive heterotopic neurons [a]		mMCD with oligodendroglial hyperplasia in epilepsy (MOGHE) [c]	
No definite FCD on histopathology [a]	Abnormality of cortical organization remains ambiguous and histopathological findings not compatible with FCDI, II or III [d]			

https://pmc.ncbi.nlm.nih.gov/articles/PMC9545778/

FCD type Ia



FCD type Ib



The FCD type Ic is the combination of both variants. Even the neuropathologic experts doubts whether it is real entity or not.

1. FCD Type I - Architectural Abnormalities

This type is about the layers of the brain cortex being disorganized, but the brain cells (neurons) themselves look normal.

- Ia: Characterized by "microcolumns" where neurons are lined up in abnormal vertical columns.
- Ib: The horizontal layers of the cortex are blurred or fused together.
- Ic: A combination of both vertical and horizontal layering problems.

Added / revised in the official lecture

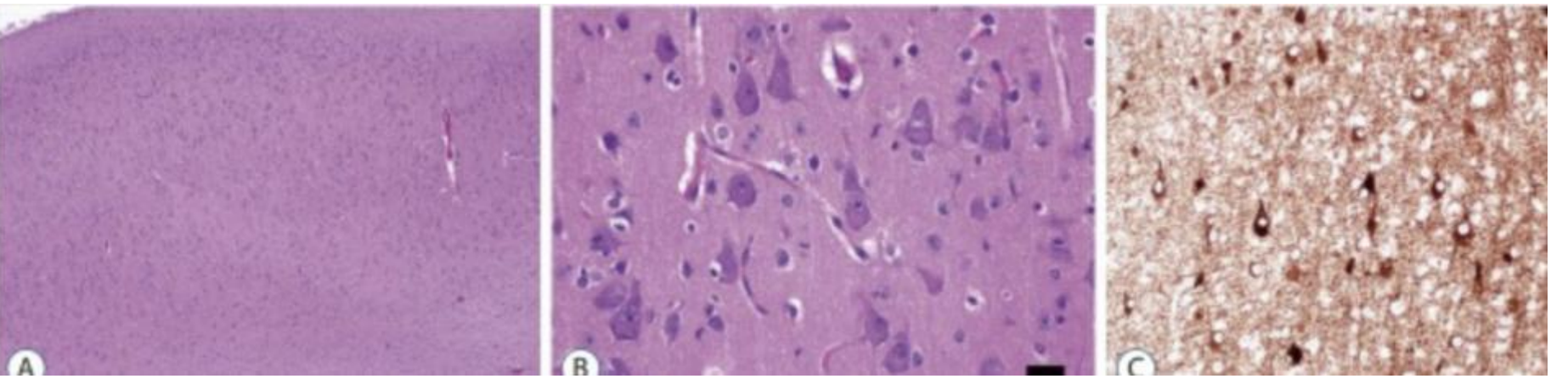
FCD type I - revised descriptions

FCD type Ia: Abundant microcolumns

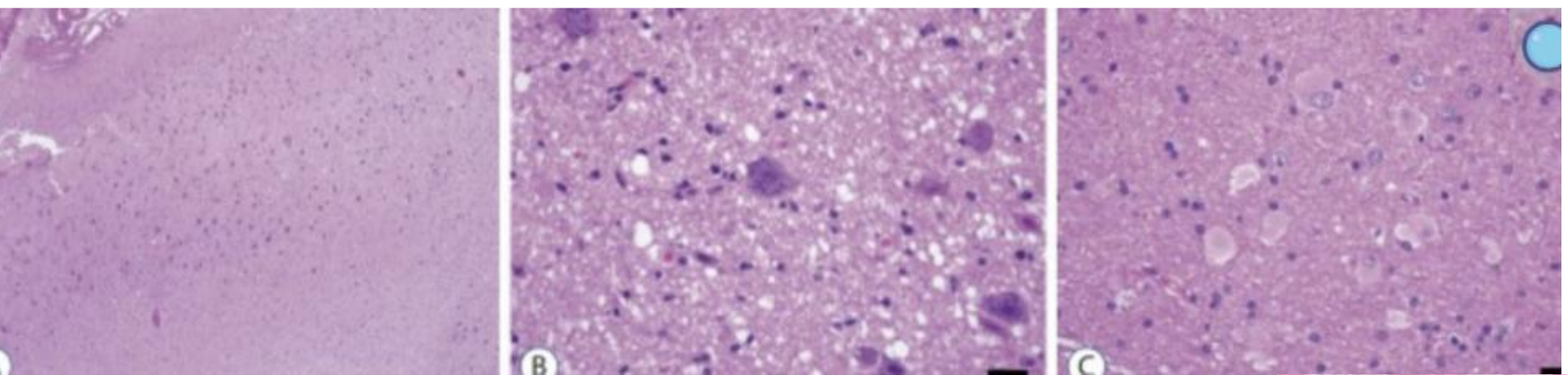
FCD type Ib: abnormal layering

Revised type Ic caption: The FCD type Ic is the combination of both variants: vertical and horizontal abnormalities

FCD type IIa

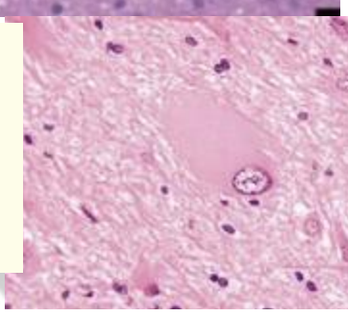


FCD type IIb



2. FCD Type II - Cytoarchitectural Abnormalities (More Severe)
This type has both disorganized layers and abnormal, dysmorphic (misshapen) neurons.

- IIa: Features dysmorphic neurons.
- IIb: Features dysmorphic neurons plus "balloon cells." These are large, abnormal cells that are a hallmark of this type.



Added / revised in the official lecture

FCD type II - added image descriptions

FCD type IIa: cortical dyslamination (derangement of normal neocortical six-layers with large dysmorphic neurons with coarse Nissl substances. The size of dysmorphic neuron cytoplasm is more than 20 µm)

FCD type IIb: cortical dyslamination with large dysmorphic neurons and balloon cells showing eosinophilic homogenous cytoplasm

Focal Cortical Dysplasia

FCDIII [a]	FCDIIIa cortical dyslamination associated with hippocampal sclerosis	FCDIIIb cortical dyslamination adjacent to brain tumor	FCDIIIc cortical dyslamination adjacent to vascular malformation	FCDIIId cortical dyslamination adjacent to lesion acquired during early life, e.g. stroke

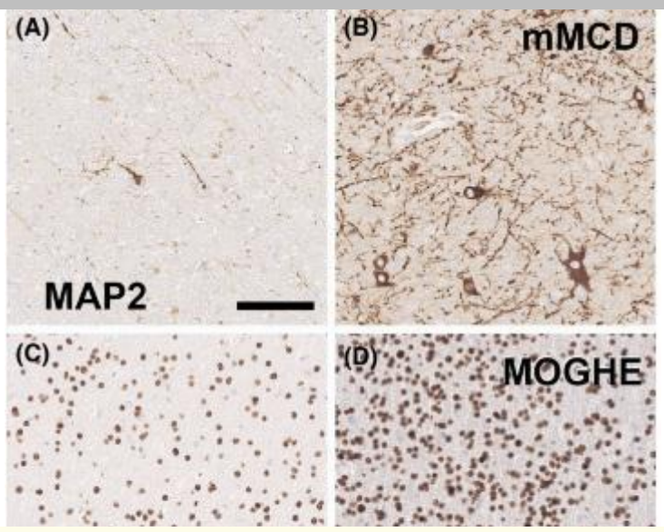
3. FCD Type III - Associated with Another Principal Lesion

In this type, the cortical dysplasia occurs alongside another, primary brain abnormality.

- IIIa: Associated with Hippocampal Sclerosis (scarring in the Hippocampus -Memory Center-).
- IIIb: The dysplasia is right next to a brain tumor.
- IIIc: The dysplasia is adjacent to a vascular malformation (an abnormal tangle of blood vessels).
- IIId: The dysplasia is next to a lesion from an early life injury, like a stroke or trauma.

Focal Cortical Dysplasia

White Matter [a]	mMCD [b] with excessive heterotopic neurons [a]	mMCD with oligodendroglial hyperplasia in epilepsy (MOGHE) [c]
No definite FCD on histopathology [a]	Abnormality of cortical organization remains ambiguous and histopathological findings not compatible with FCDI, II or III [d]	



4. mMCD "Mild Malformation of Cortical Development."
This category includes milder malformations.

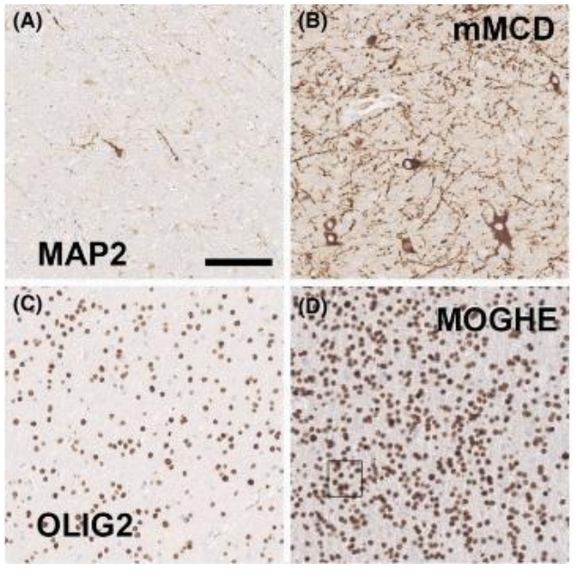
- one type is: Excess of neurons in the white matter (the brain's deeper wiring).
- Another is: MOGHE (mMCD with oligodendroglial hyperplasia in epilepsy), which involves an overgrowth of a specific type of brain support cell.
- There is also a category for cases where the tissue abnormality is clear but doesn't fit neatly into the FCD I, II, or III groups.

Added / revised in the official lecture

Mild malformation of cortical development

Upper image row (MAP2 / mMCD):
Mild malformations of cortical development with heterotopic neurons

Lower image row (OLIG2 / MOGHE):
Mild malformations of cortical development with oligodendroglial hyperplasia



Full image from the official revision

Hippocampal Sclerosis (Mesial Temporal Sclerosis)

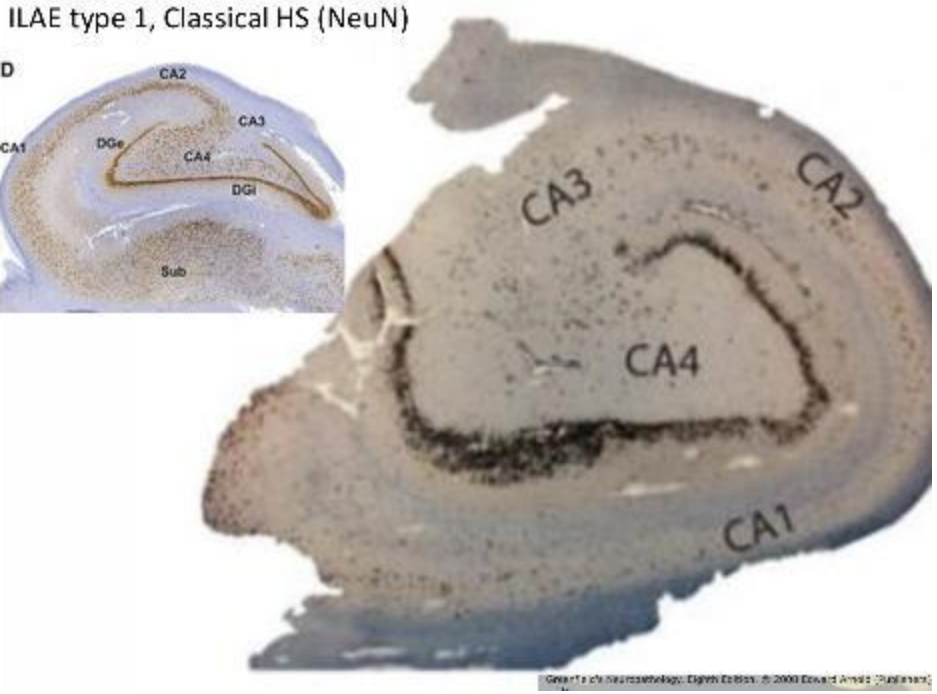
- The most common pathological feature of temporal lobe epilepsy (TLE)
- Pathologic examination reveals neuronal loss in the hippocampus, predominantly involving the hilar region, CA1, CA3, CA4, and dentate gyrus, with relative sparing of the CA2 region with giosis
- ILAE recognizes three pathological subtypes

Subtypes added in the official revision

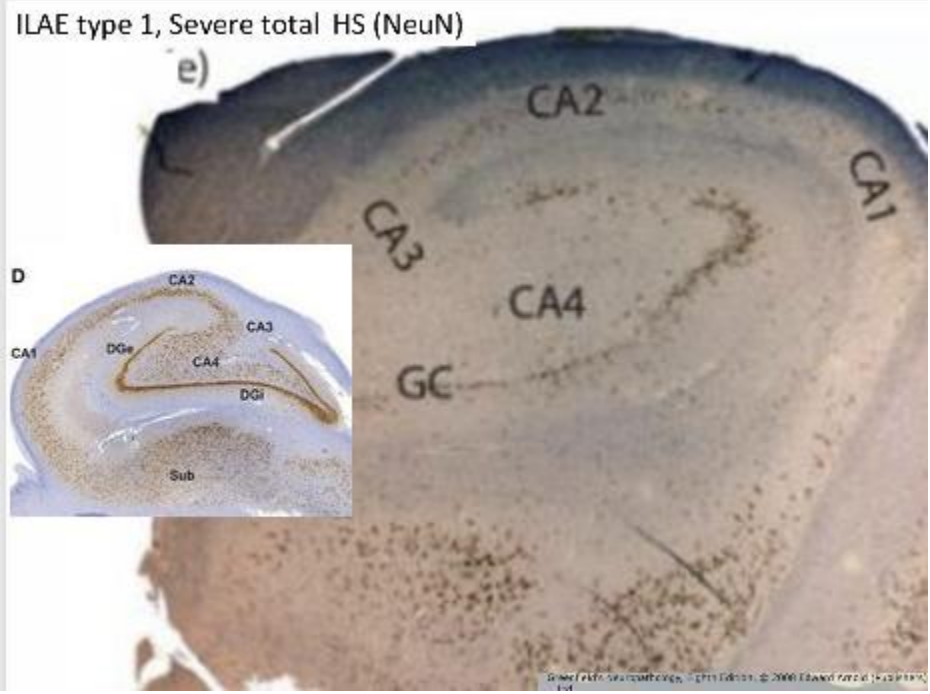
- 1 classical: neuronal loss of all CA sectors 1-4
- 1 severe: severe neuronal loss of all CA sector 1-4
- 2: neuronal loss of CA1
- 3: neuronal loss of CA4

Official revision spelling: "gliosis" (in the paragraph above).

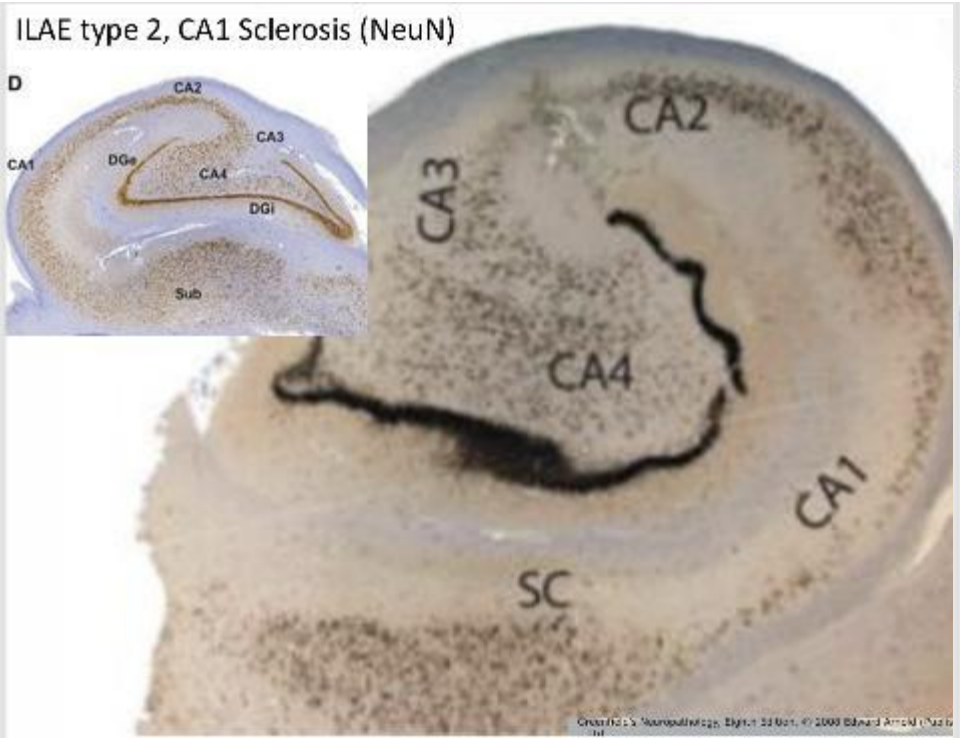
ILAE type 1, Classical HS (NeuN)



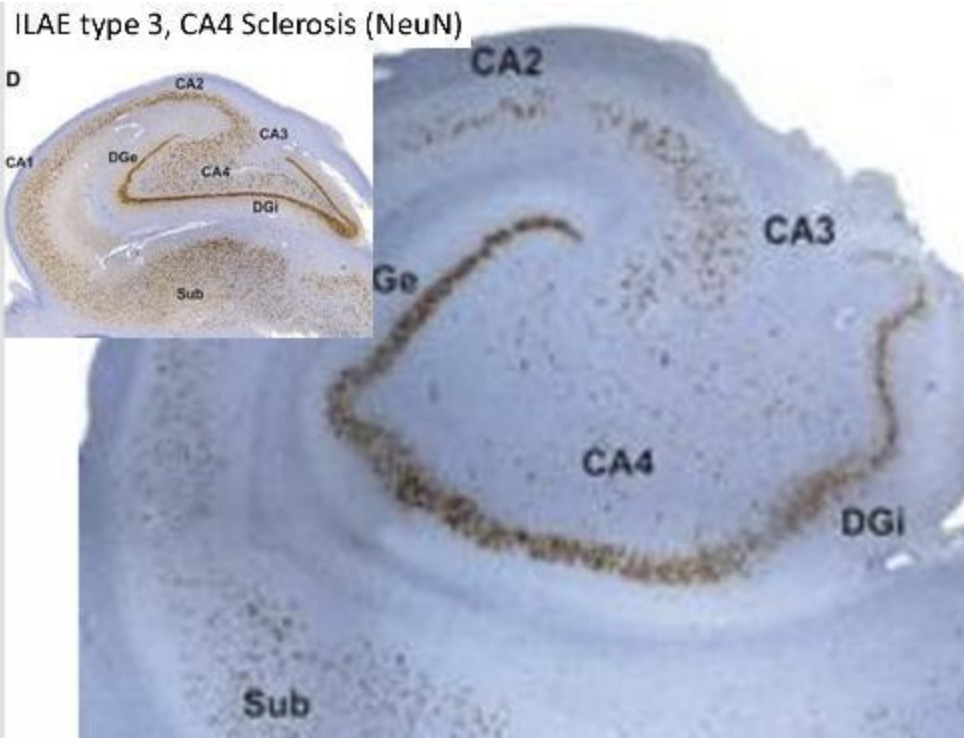
ILAE type 1, Severe total HS (NeuN)



ILAE type 2, CA1 Sclerosis (NeuN)



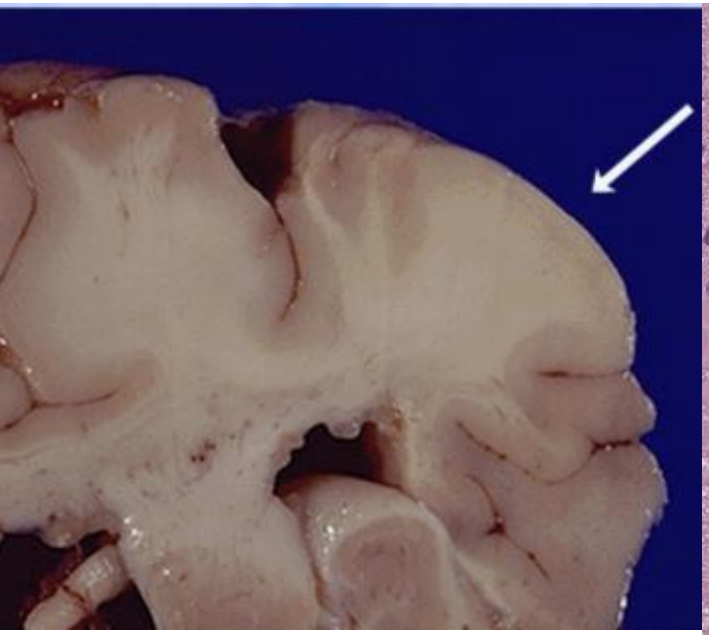
ILAE type 3, CA4 Sclerosis (NeuN)



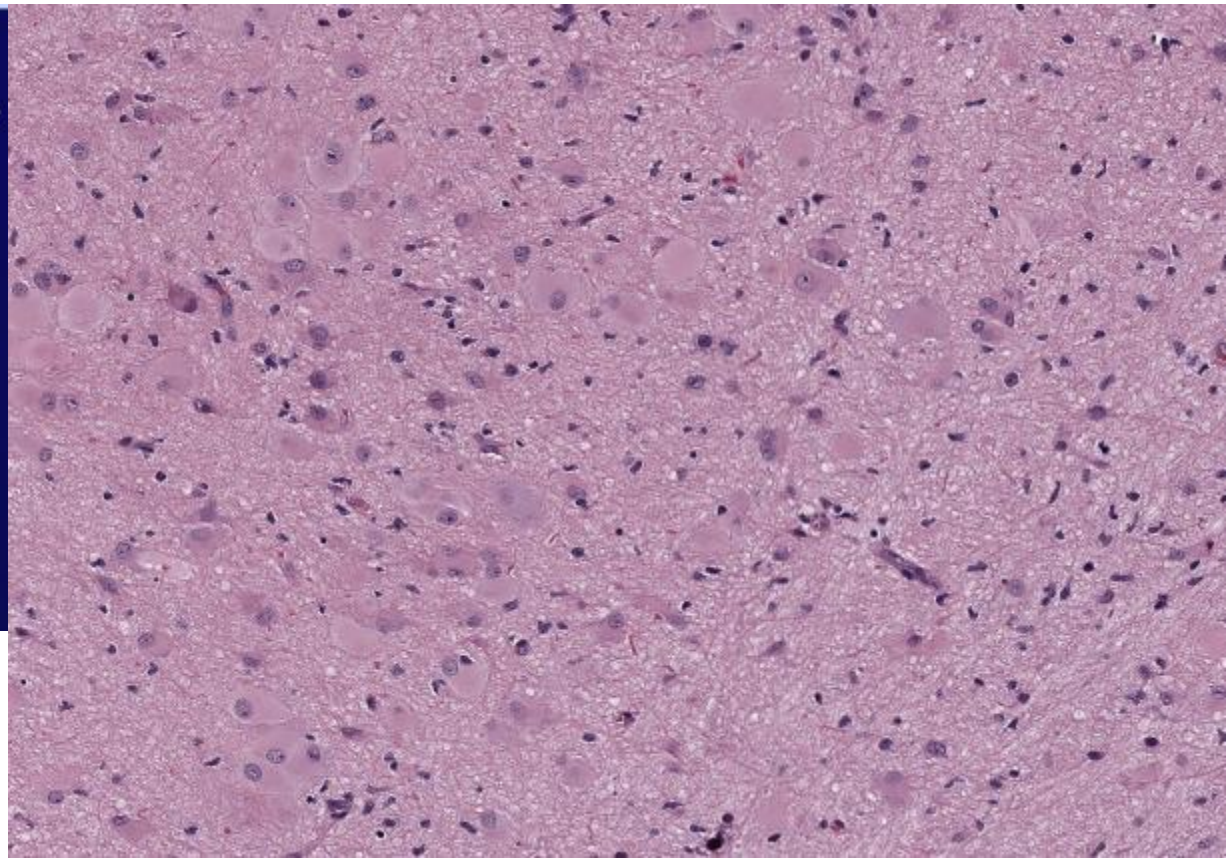
Tuberous Sclerosis

- Autosomal dominant disorder
- Characterized by the development of hamartomas and benign neoplasms involving the brain and other tissues
  - CNS: hamartomas and tumour
    - Subependymal giant cell astrocytoma (SEGA)
    - Subependymal hamartomas
    - Retinal hamartomas
    - Cortical tubers: epilepsy, difficult to control with antiepileptic drugs
  - Extracerebral lesions include: multiple cysts

Tuberous Sclerosis



https://biology.kenyon.edu/slonc/bio38/howell/tsc.htm



https://tumourclassification.iarc.who.int/Viewer/Index2?fid=17643

Tumours

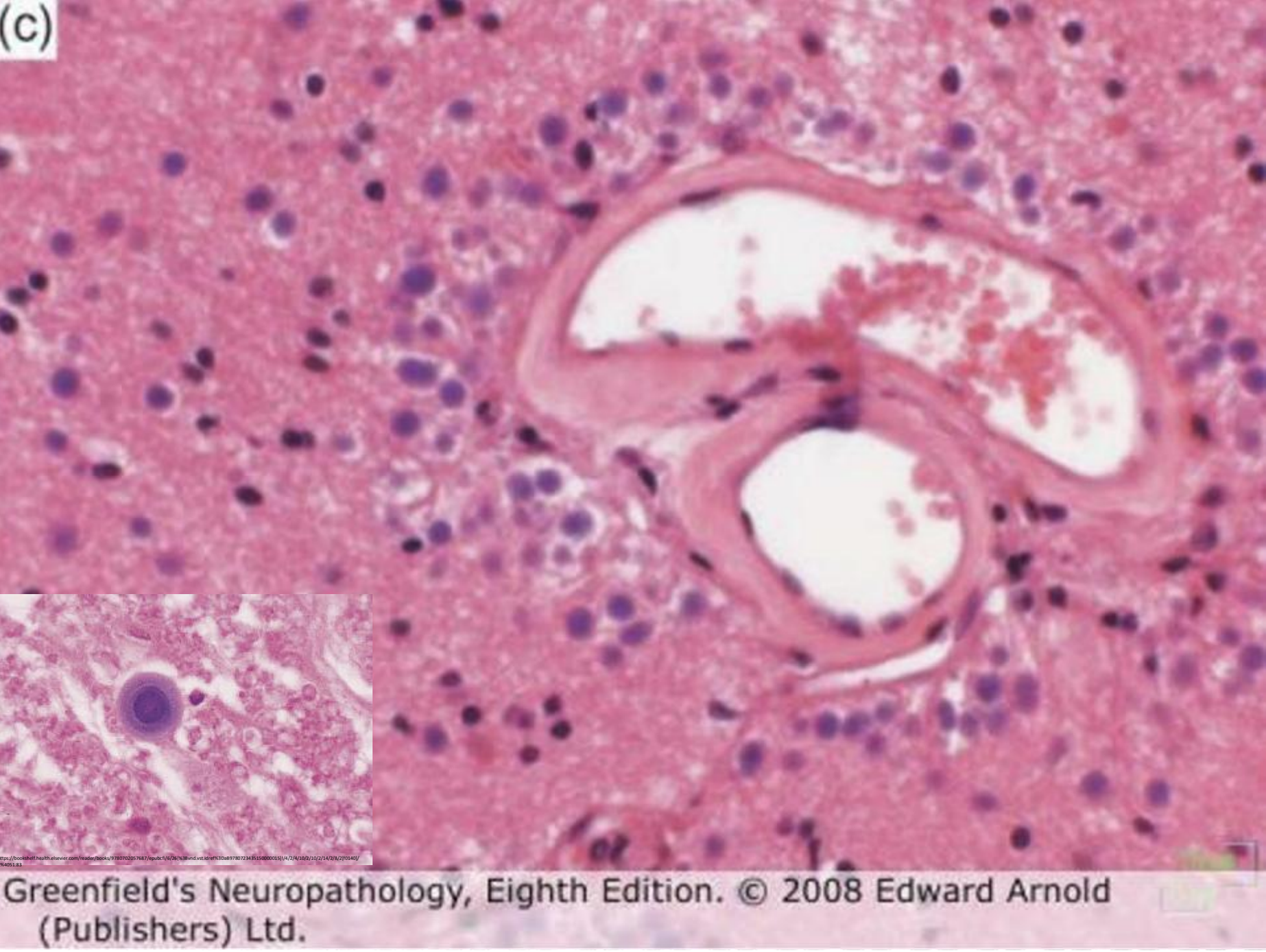
- Certain CNS tumors are associated with drug-resistant epilepsy
  - Dysembryoplastic neuroepithelial tumour (DNT)
  - Gangliogliomas/gangliocytoma
  - Pleomorphic xanthoastrocytoma(PXA)
  - Angiocentric glioma

Secondary Pathology in Epilepsy

Subpial gliosis



Greenfield's Neuropathology, Eighth Edition. © 2008 Edward Arnold (Publishers) Ltd.



Greenfield's Neuropathology, Eighth Edition. © 2008 Edward Arnold (Publishers) Ltd.

Added / revised in the official lecture

Corpora Amylacea

- Spherical inclusions
- Predominantly in astrocyte processes
  - Subpial area
  - Around blood vessels
- Consist largely of polyglucosans

amakhotani@uqu.edu.sa

